1ZV60 Methodology for IE Research – Summary 2026-2027

Purpose of this document: study summary of the 2026-2027 course material (Lectures 1.1, 1.2, 2.1, 3.2, 4.1, 5.1, 6.1, 6.2, 7.3/7.4) at TU/e, Industrial Engineering & Innovation Sciences (ITEM group). The exam is multiple choice (MPC); it covers the lecture slides/handouts and the listed chapters of Bell, Bryman & Harley (2022), *Business Research Methods*. Tip: the course model answers use exact lecture terminology, so learn definitions word for word.

Course structure

Block	Weeks	Content
General introduction	Week 1	Research methods, theory, philosophy, cycles, ethics
Qualitative methods	Weeks 2-4	Qualitative design, sampling, data collection, analysis
Quantitative methods	Weeks 4-6	Quantitative designs, questionnaires, measurement, sampling, secondary data
Problem-solving research	Weeks 7-8	Cause-effect diagrams, demarcating problems, regulative cycle

Example exam questions (MPC format)

- **Q1 – Population.** Hilton estimates 10% of 2,500,000 Hilton Honors Club (HHC) members are brand loyal. The population in this research = **guests registered as HHC members** (not only the loyal 90%, not all travellers, not all Hilton guests).
- **Q2 – Secondary data analysis** = using information that has **already been collected and recorded by someone else for other purposes**.
- **Q3 – Unit of analysis.** Researcher asks engineers in teams at a Swiss company how much time they spend on product creation and links this to their work motivation. The unit of analysis = **individual engineers at this company** (the level at which the hypothesis is tested – not the teams or the company).
- **Q4 – Problem-solving cycle sequence** = **Define problem, diagnose, design, intervene, evaluate**.

Lecture 1.1 – Introduction to Research Methods

Dr. Paul Moritz Wiegmann. Exam material: lecture slides + Bell et al. (2022) Ch. 1 (nature and process of business research), Ch. 2 (business research strategies), Ch. 6 (ethics), (Ch. 26 breaking down the quantitative/qualitative divide).

Why good methodology matters

- The industrial engineer is a **"doctor for businesses"**: both industrial engineering and medicine are fields based on multidisciplinary academic knowledge. Object of study: business (IE) vs. human (medicine). Goal: make the business healthy vs. make people healthy. Both need a **systematic approach to building and applying knowledge**.

Method = "a systematic established procedure for approaching something". **Methodology** = "a system of methods used in a particular area of study" (Oxford dictionary).

- Why YOU need methods: make informed choices in your projects (e.g. BEP); plan and set up your project; apply methods to get good results and avoid pitfalls; assess research critically and understand its limitations; learn transferable skills; gain an academic mindset.

The role of theory

- **Theory = a set of general rules.** Theory (the rule) sits above the line; **empirical evidence** (cause → effect) below it.
- Example theory: **Technology Acceptance Model (TAM)** (Davis, Bagozzi & Warshaw, 1989): external variables → perceived usefulness and perceived ease-of-use → attitude towards using → behavioural intention of use → actual system use (perceived ease-of-use also affects perceived usefulness; perceived usefulness also directly affects behavioural intention).

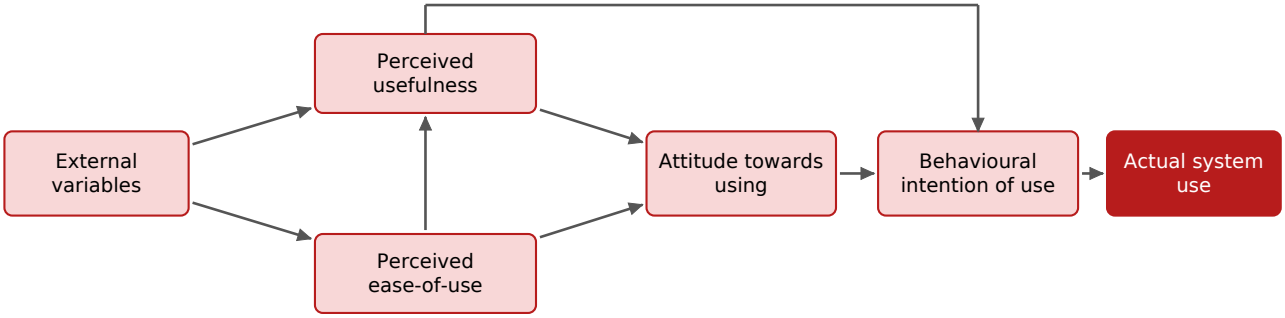


Figure – Technology Acceptance Model (Davis, Bagozzi & Warshaw, 1989): external variables → perceived usefulness and perceived ease-of-use → attitude towards using → behavioural intention of use → actual system use.

Key theory-related terminology (Dul & Hak, 2008)

Term	Definition	TAM example
Domain	All instances to which the theory is expected to apply	All potential users of new computer technologies
Population	A sub-set of the domain where the theory can be developed and/or tested	All potential users of smartwatches in the Netherlands
Sample	The instances from the population that are studied	Potential Dutch smartwatch users reached via e-mail
Unit of analysis	The instance to which the theory applies	A potential user of a new computer technology

- Nesting: domain ⊃ population ⊃ sample.

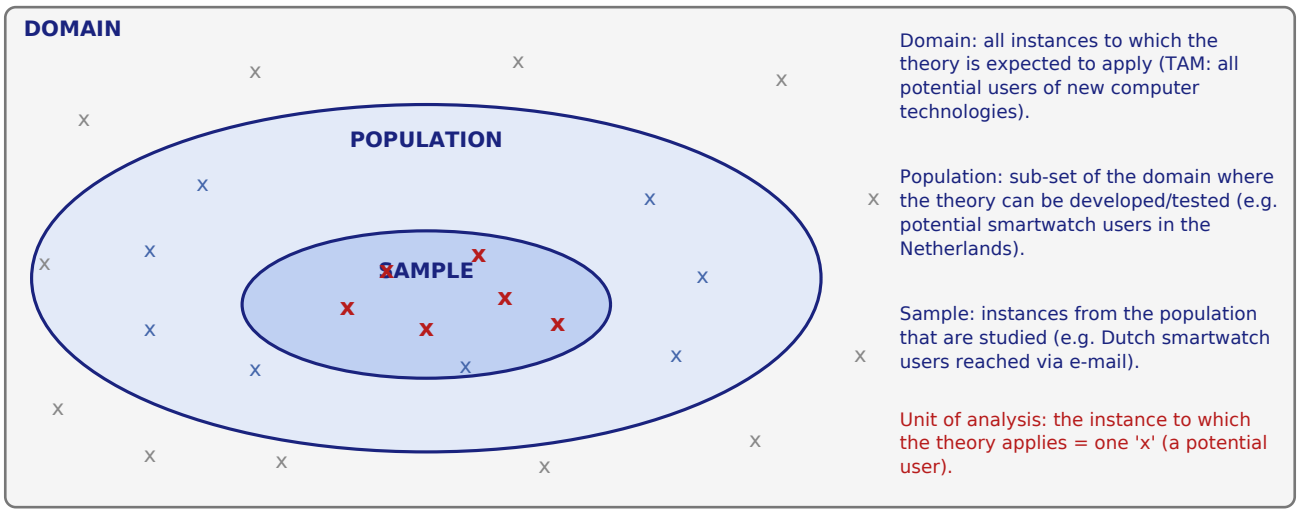


Figure – Domain ⊃ population ⊃ sample; the unit of analysis is the instance to which the theory applies (one potential user in the TAM example).

Theoretical reasoning – three ways to "cross the line" between theory and empirical evidence

Reasoning	Given	Find	Example
Induction	A cause and an effect	The rule	Observation: store managers deviate from the amount recommended by the forecasting system; effect: more food ordered than can be sold → *possible* rule: when managers deviate from the recommended amount, food is wasted
Deduction	The rule and the cause	The (expected) effect	Rule: the more a company spends on advertising, the more it sells; observation: company X increased advertising → *expected* effect: sales increase
Abduction	A rule and an effect	A (possible) cause	Rule: insufficient information to supplier → raw materials delivered too late; observation: raw materials late → *possible* cause: insufficient information was provided

Philosophical foundations – how can theories be supported?

Depends on your views about **ontology** (what kinds of objects exist in the "social world"?) and **epistemology** (what can be considered "acceptable knowledge"?).

Ontology – how social (organisational) phenomena and their meanings exist

Objectivism	Constructivism (constructionism)
Phenomena and their meanings exist independently of actors. Implication: phenomena are objectively observable (e.g. organisational hierarchies, rules and regulations, procedures).	Phenomena and their meanings are constantly accomplished by actors. Implications: phenomena are produced through interaction and constantly revised; researchers' accounts of the world are constructions; knowledge is subject to change.

Epistemology – what knowledge is and how it is produced

Positivism	Interpretivism
Apply a "natural science approach" to **explain** social phenomena. Phenomena can be observed/measured objectively; theory provides generally applicable "laws" that can be tested deductively; new knowledge is derived from induction based on facts; researchers are objective, value-free observers.	Attempt to **understand** social phenomena. Phenomena have subjective meaning (to researchers and social actors); researchers try to understand this meaning for the actors they study; the researcher's interpretation is influenced by own experience; being objective and value-free is neither possible nor desirable.

- **Implications for methods:** appropriate research methods depend on ontological and epistemological foundations (research methodology tree: ontology = roots, epistemology, methodology = how we should best collect data, methods = exact details of data collection). **Most research at TU/e and in this course is based on objectivist/positivist foundations.**

Two research cycles

	Empirical cycle	Problem-solving cycle
Question	**How to explain it?**	**How to make it work?**
Phases	Observation → Induction (developing theory) → Deduction (generating hypotheses) → Testing (of hypotheses) → Evaluation	Problem definition → Analysis and diagnosis → Solution design → Intervention → Evaluation and learning (all around a central "problem mess")
Used for	Generic business problems many businesses encounter; relationships between variables; develop and empirically test theories; aim = generalisable insights usable across contexts	Performance-related business issues (a business process not meeting a defined performance level, e.g. costs, quality, timeliness); **context specific**
Use when	"How things work" is not understood	"How things are done" needs to be improved

- Examples from the slides: "Does more spending on advertising lead to more sales?" → empirical cycle. "How can we reduce food waste at the local Albert Heijn?" and "Too many trains in the Netherlands are delayed" → problem-solving cycle.
- **The two cycles complement each other:** empirical research provides **theoretical foundations for a solution**; problem-solving research provides **relevance for theory development**.

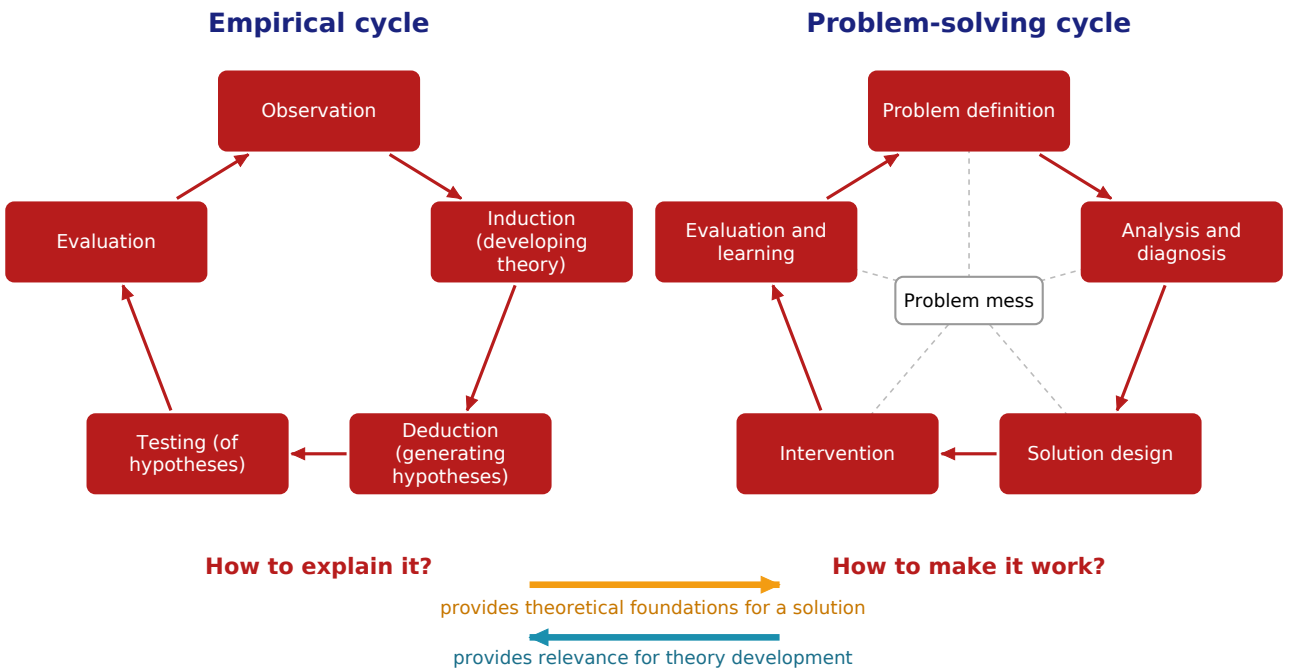


Figure – Empirical cycle (how to explain it?) and problem-solving cycle (how to make it work?) and how they complement each other.

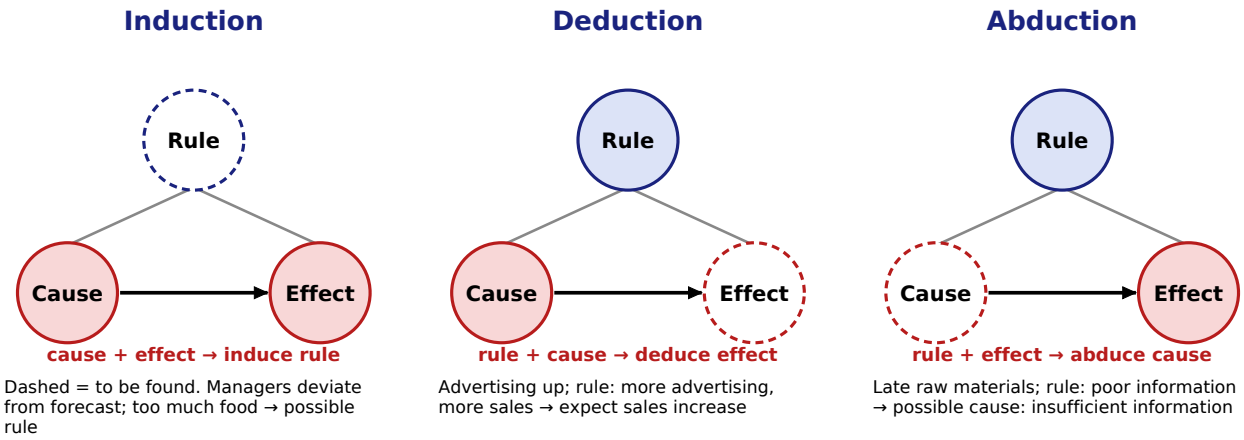


Figure – Induction (cause + effect → rule), deduction (rule + cause → effect) and abduction (rule + effect → cause); the dashed circle is what you derive.

Research ethics

- Ethics do not go without saying. Case: **Dirk Smeesters** (marketing professor, Erasmus University Rotterdam): an anonymous researcher found inconsistencies in published data; investigation showed "massaging of data"; papers retracted, he left EUR. Similar cases: Diederik Stapel (Tilburg), Ulrich Lichtenthaler (Mannheim).
- Ethical issues you will meet: protecting participants' privacy while keeping results traceable; dealing with missing data; dropping an observation; results that differ from what the client expected.
- Not always clear-cut: e.g. studying regulatory oversight and workplace safety – what to do with observed safety violations?
- Guidelines: TU/e Code of Scientific Conduct; Bell et al. checklist (2019, p.135); Martin (2016): "Would I feel embarrassed if my activities came to light and would I therefore object if I was publicly named?"

Learning goals

Explain why good methodology matters; use induction, deduction and abduction; articulate ontological and epistemological views; identify which cycle fits a problem; identify ethical dilemmas.

Lecture 1.2 – Developing Research Questions, Writing up Research, Introduction to Assignment 1

Dr. Paul Moritz Wiegmann, 3 September 2026 (the slides themselves are labelled "Lecture 1.4"). Exam material: lecture slides, the description of Assignment 1, Bell et al. (2022) Ch. 4 (planning a research project and developing research questions), Ch. 5 (reviewing the literature), Ch. 7 (writing up business research).

Developing good research questions

- **Why good RQs matter:** they provide **focus**; are a **point of reference** throughout the project (methodological decisions, study design); allow **evaluation of the results** (have the aims been met?).
- **Sources of research questions** (Bell et al.): intellectual puzzles and contradictions, existing literature, replication, structures and functions, opposition to existing ideas/findings, social problems, counter-intuitive observations, empirical observations, application of new methods and theories, new developments and trends, personal experience, sponsors (internship supervisors, lecturers).
- Getting to your RQ: from a general area of interest, narrow down to a research question (Bell et al.).

Criteria for assessing RQs: a good RQ is **clear**; **researchable**; **connected with established theory and research**; **logically and conceptually linked** to the other RQs (if more than one); (offers potential to contribute to knowledge); **neither too broad nor too narrow**.

Reviewing literature

- **Why a literature review:** overview of what is already known; learn from others' mistakes; learn about different theoretical and methodological approaches; helps to develop an **analytical framework**; may reveal insights/concepts you had not thought of; may suggest further RQs. In short: what do we (not) know already, why is the topic relevant, what should be done to get more relevant knowledge.
- **Finding journal articles:** (1) keyword search (Web of Science, Scopus, Google Scholar; keep track of keywords), (2) reading relevant journals (those that frequently appear in results), (3) following citations: **backward citations** (what the sources you found cite) and **forward citations** (who cites them – Web of Science/Google Scholar).
- **Assessing quality:** not all journals are equal; **avoid predatory journals**; check reputable lists (ISI Journal Citation Reports/impact factor, FT50, ERIM list); citation count (few citations do NOT mean low quality); above all use your own judgement and be critical.
- **Other literature:** books (academic books vs. textbooks; check author and publisher); professional journals (what happens in practice, cannot replace academic sources); working/conference papers (smaller time lag, often not peer reviewed; SSRN); press articles (illustrative examples); websites (be critical; record date of access and complete URL).

Types of literature review (Bell et al., 2022)

Systematic review	Narrative / integrative review
Positivist (?); "quantity" focus; theory informs the search; synthesis is the research goal	Interpretivist (?); "quality" focus; theory emerges from the search; identifying the discourse is the research goal

- **Structure of a review:** three alternatives – **concept-centric** (recommended), author-centric (not recommended), paper-centric (not recommended). The review must support the argumentation of the entire report.

Writing up research

- Follow the requirements and instructions; start early; structure your writing; be persuasive; avoid discriminatory language; use your resources and ask feedback frequently (not only from lecturers).

Assignment 1 (qualitative research project)

- **Purpose:** learning research methods requires learning-by-doing – you conduct your own qualitative research project. **Topic: sustainable innovation** – it only helps if it is adopted. What causes and/or hinders consumers to adopt these innovations? **Your job: develop a theoretical (conceptual) model to answer this question.**
- **Steps:** review literature and define the RQ → design the empirical research → collect data (**interviews and documents**) → analyse data → develop a conceptual model → write up the report. (1 RQ is enough; some parts are already taken care of by the teaching team.)
- **Dates:** 22 September advice clinic; **4 October deadline** (hand in via Canvas). **Teams of 6**; input needed from all members at all stages.
- **Individual peer grading:** each student evaluates all group members' contributions; group grades may be adjusted up or down per student.
- **Report form:** academic paper + appendix. Main report: Introduction & RQ, Methodology, Findings, Discussion, References. Appendices: reflection on process, statement about interviews, **statement on the use of AI**, interview guideline, excerpts from interview transcripts, codebook.
- **Grading (8 criteria):** literature review and research question 10%, research design 20%, interview data collection 15%, document data collection 10%, data analysis 20%, conceptual model 10%, writing 5%, reflection 10%. Final grade = weighted average of individual grades.
- **Use of AI tools:** see TU/e Information Literacy Hub; **report whether and how you used AI**; do NOT use AI for fabricating data.

Lecture 2.1 – Qualitative Research Design, Sampling and Data Collection

Dr. Paul Moritz Wiegmann, 7 September 2026. Exam material: lecture slides, description of Assignment 1, Bell et al. (2022) Ch. 17 (nature of qualitative research), Ch. 18 (sampling in qualitative research), (Ch. 19 ethnography and participant observation), Ch. 20 (interviewing in qualitative research), (Ch. 21 focus groups), Ch. 23 (documentary data).

When to use qualitative research

- **Trade-off in empirical research:** quantitative research → **breadth (generalisability)**; qualitative research → **depth, flexibility, context**.
- **In the empirical cycle** qualitative approaches are suitable for: **induction** (grounded theory, exploratory studies in a new field), matching theoretical explanations to qualitative observations, and **testing** through case studies of most/least likely cases.
- **In the problem-solving cycle:** studying the existing situation, qualitative analysis of similar solutions, getting feedback on solution ideas, tracking progress of the intervention, collecting feedback from users.
- Views from the literature: **Yin (2009):** answer "how/why" questions when you have little control over events and study "a contemporary phenomenon in a real-life context"; **Creswell (2006):** when no/insufficient theory is available and/or in-depth understanding is needed; **Jaspers (2007):** process tracing, most/least likely cases, pathway cases; **Jansen (2010):** capturing diversity in a population; **Flick (2009):** when the object of study is too complex to be studied otherwise.
- **Types of qualitative research** (Creswell, 2014; Bell et al.): narrative research, phenomenological research, **grounded theory**, ethnography, **case studies**, many mixed-methods flavours. Researchers often develop their own approaches; often (not always) constructivist/interpretivist.

Case study	Grounded theory
Aim: in-depth analysis of case(s); theory building (or testing) based on the analysis; common methods (Eisenhardt 1989; Yin 2009) suit the positivist paradigm	Aim: "discover" a theory from the data; positivist and interpretivist approaches exist

- Similar approaches to case selection and data collection (but different vocabularies); **different approaches to data analysis**.

Qualitative study designs and sampling cases

- **Unit of analysis** usually follows from the research question; it is the first question when designing a study; the case(s) you select must be instances of your unit of analysis. Examples: company, project, team, individual, process, product. (Slide quiz: "How do problems in families affect students' study progress?" → individual student; "What determines effectiveness of advertising campaigns?" → advertising campaign; "How should the NPD process be adapted to enable re-usable products?" → the NPD process.)

Single vs. multiple case designs

- **Single case** (grounded-theory approaches often focus on one case). **Five rationales (Yin, 2009):** (1) **critical case** (often theory testing), (2) **extreme/unique case**, (3) **representative/typical case**, (4) **revelatory case**, (5) **longitudinal case**. Additional reason: practical constraints.
- **Multiple case design:** seen by some (Eisenhardt, 1989) as the default; enables **replication**; uses within-case and cross-case analysis; when building theory it gives a first test. **Literal replication:** sample cases where you expect **similar** findings. **Theoretical replication:** sample cases where you expect **different** findings (Yin, 2009).

Two levels of sampling

1. Selecting case(s) for the study; 2. selecting data sources within the case(s).

- **Do not use random sampling for cases; use purposive sampling** (choose cases from the population based on pre-defined criteria). Random sampling = equal chance for any case from the population.

- **Types of purposive sampling:** extreme cases, typical cases, critical cases, maximum variation, criterion sampling, theoretical sampling, snowball sampling, **opportunistic (convenience) sampling**, stratified purposive sampling.
- **Finding a good case:** define unit of analysis; what key information is needed per case (e.g. detailed timeline, measure of success); practical requirements (access to key interviewees/documents, not too long ago, geographical distance); decide what is essential; share aims upfront with case contacts.

Data collection

- **Sources:** nearly everything can be evidence: interviews, internal documentation, e-mails, company reports, media articles, videos, **quantitative data**, personal observations, physical artefacts.
- **Sampling data sources (Jansen, 2010):** maximise diversity, capture all perspectives; start with a small sample → initial analysis → decide on additional sources based on it → define a **stopping rule (ideally saturation)**.

Quality criteria in business research (Bell et al.)

Criterion	Question
Reliability	Would repeating the study produce consistent results?
Replicability	Could somebody else get the same results? (sometimes included under reliability)
Validity	Do the conclusions accurately reflect what is actually going on?

- Slide examples (student absence study): asking three lecturers why students skip = questionable validity (wrong perspective); students repeatedly saying absent students are "lazy" = consistent (reliable) but possibly not valid; asking the first three students you meet = convenience, neither reliable nor valid; asking a **diverse group** of students who have skipped = valid and better reliable.

Principles for validity and reliability

- **Triangulation** (Yin, 2009): converge multiple sources of evidence on the same finding.
- **Saturation:** collecting additional data does not reveal new information. Cover diversity of the population (no absolute certainty possible); establish **and document** rules of thumb (e.g. stop after the last 5 new data sources revealed no new insights); strongly linked to data analysis.
- **Chain of evidence** (Yin, 2009): traceable link from research question → protocol → evidence → database → report.
- **Research design protocol:** record all procedures and reasoning in one document; update it regularly; readers (including team members) should be able to perform the same study.

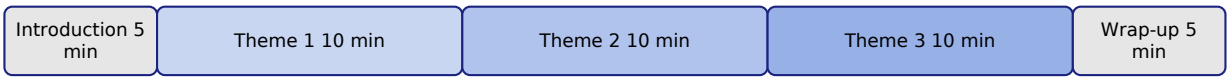
Interviewing

Interview types

Type	Description
Unstructured	Start with a very broad question/theme ("Tell me about problem XY"); interviewee determines the direction
Semi-structured	Interview guideline with a few broad topics/questions; **5-10 questions (max) for a 1-hour interview**; **most common approach for IE**
Structured	Detailed questionnaire with specific questions

Before the interview

- Preparing the guideline: avoid jargon; **avoid leading questions**; use existing data and theory (update guideline between interviews); logical order; prioritise questions given time (with back-up questions); have it checked by someone else.
- **Typical structure:** 5 min introduction; 10 min theme 1; 10 min theme 2; 10 min theme 3; 5 min wrap-up; informal after-talk. Within each theme start with **open questions**, possibly more closed questions towards the end, give a summary before moving to the next theme; be flexible.



Informal after-talk follows. Within each theme: start with open questions, more closed questions towards the end, summarise before moving to the next theme.

Figure – Typical semi-structured interview (about 40 minutes of themes plus informal after-talk).

- Question quality: good = open, explanatory, concrete ("Please explain the criteria you use to decide how much stock to order"; "Think about the last time you had capacity issues – what happened?"). Bad = **leading or loaded** ("Why do you always order more stock than required?"; "Don't you agree the process leads to ordering too much?"; assumes a problem: "Why is there too much food waste at your shop?").
- Practical: setting with little distraction; contact interviewee (purpose, confidentiality terms, format and duration, questions/contact details, guideline in advance?); recording device with permission, **always take notes anyway**; print-outs of the guideline.

During the interview

- **Mistakes:** no explanation of purpose, no rapport, no thinking time, showing no interest, rushing, leading questions, interviewer answering questions, not paying attention, impolite.
- **Good practice:** introduce purpose and methods; build trust and rapport; build confidence; keep neutral; do not push; give time to think; allow the interviewee to correct himself/herself; **interviewee does most of the talking**.
- **Active listening (probing, summarising):** "Could you tell me more about that?", "What do you mean exactly by …?", "Can you give some examples?", "Do I understand correctly that …?", "Mhm…", **mirroring** (repeat the last few words), **silence**.
- **Ethical and legal aspects:** state the research purpose and what you do with the information; interviewee may stop at any time or skip any question; clear agreements about **confidentiality and anonymity**; **explicit permission to record, and start recording only afterwards**; you are bound by the **GDPR** and the TU/e Code of Scientific Conduct.

After the interview

- Take field notes as soon as possible; transcribe the interview; validate responses.

What next? Iterate between data collection and analysis

- Qualitative research is iterative: plan project → collect data → analyse data → (back to literature/theory), e.g. the Scrum study (exploration with interviews and participant observation → formal semi-structured interviews → open and axial coding, iterating between theory and data).

Lecture 3.2 – Qualitative Data Analysis and Reporting

Dr. S.A.M. Dolmans (ITEM group). Exam material: lecture slides + Bell et al. Ch. 22 (language in qualitative research), Ch. 23 (documents as sources of data), Ch. 24 (qualitative data analysis), (Ch. 25 computer-assisted qualitative data analysis).

Why qualitative analysis is challenging and rewarding

- Fewer standardised rules and practices for analysis and reporting → more uncertainty but also more room for agency and creativity. **No software does the analysis for you**: you do the bulk of the analysis and sensemaking; findings are a direct result of your own "processing power". Software only helps with organising data, coding, and generating insights.
- **Key question:** how to make sense of large amounts of qualitative information **in a valid and reliable manner**?

Coding

Code (Miles & Huberman, 1994, p.56) = a tag or label for assigning **units of meaning** to descriptive or inferential information compiled during a study; attached to "chunks" of varying size – words, phrases, sentences or whole paragraphs. Often thought of as putting comparable data segments into "buckets" (Locke et al., 2020).

- Questions to identify meaning: What is happening here? What is someone (not) doing? What is this an example of?
- **Something is relevant to code when:** it relates to the research question; it is mentioned frequently or repeated; it surprises you; the interviewee explicitly states it is important; you read about something similar in articles; it reminds you of a theory or concept; any other reason you find relevant.

Two "extreme" approaches to coding

	Inductive coding / grounded theory approach	Deductive coding / thematic analysis / template approach
Focus	Data-driven	Theory-driven
Interpretation	Searching for the "insiders' perspective"	Applying a theoretical perspective
Coding	Start with **open coding** and let "local theory" emerge	Coding with a **fixed, pre-defined coding scheme** (no axial/selective coding; other approaches such as counting code occurrences may be used)
Relations	Explanation building, uncovering concepts and relationships between them	Pattern matching

- The lecture focuses on the inductive approach (which fits the **induction** step of the empirical cycle).

Three stages of coding (Strauss & Corbin, 1990)

Stage	Description
1. Open coding	Breaking down, examining, comparing, conceptualising and categorising data → yields **concepts** (labels given to discrete phenomena); value depends on usefulness and frequency of occurrence
2. Axial coding	Move from many open codes to a manageable number of **categories** (higher abstraction, may subsume two or more concepts); data are put back together in new ways by making **connections between categories** (e.g. phenomenon, causal conditions, strategies)
3. Selective coding	Discover the **core categories**, identify **causal links** between categories, develop the **story line**; works towards a model or explanation of the phenomenon

- You **iterate between levels of abstraction**; saturation applies at the end. Example (Delve, COVID-19 lockdown interviews): open codes (panic attack, trouble sleeping, nervous…) → category *anxiety*; cause *lockdown loneliness*; strategies *safe in-person meetups* and *virtual hangouts*.

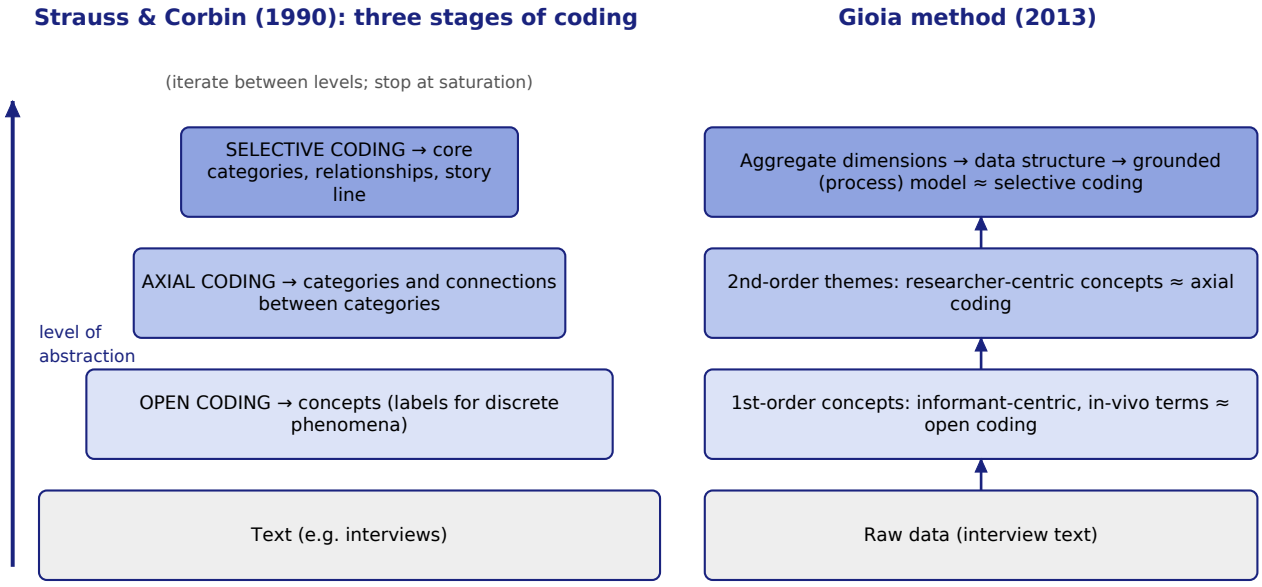


Figure – Three stages of coding (Strauss & Corbin, 1990) next to the Gioia method (1st-order concepts, 2nd-order themes, aggregate dimensions): level of abstraction increases upwards.

Rookie mistakes

- **Lack of focus / overcoding:** coding everything in a passage instead of what matters for the RQ (e.g. RQ "How do resources influence decision making?" → the relevant code is *resource slack: abundance of financial resources available*). Always go back to the RQ.
- **Uncalled quantitative sensemaking:** counting codes when it does not fit the RQ (e.g. "research methods" mentioned 4 times does not mean it is the most important course; one remark may be more insightful). Code counting is only appropriate for certain RQs (e.g. with template coding).

Codebook and constant comparison

- **Codebook (coding scheme)** contains: code, description/definition, illustrative quote(s), (optional) rules/guidelines. Aim for codes that are **mutually exclusive** and **exhaustive**.
- **Constant comparison:** each interpretation and finding is compared with existing findings as it emerges (read text completely → (re)develop codes → re-read and assign codes → read and code un-coded text).
- **Saturation:** in data collection – collecting until new data no longer illuminates the phenomenon; in analysis (theoretical saturation) – coding until there is no further point in reviewing data for fit with concepts/categories.

Memos

- **Analytic memos** capture thoughts during data collection, analysis, coding, sensemaking and conclusion drawing; not merely descriptive – they **synthesise data into higher-level analytical meanings**. Keep separate from field notes/transcripts; give clear title and date.
- Topics: RQs and reflection, codes and definitions, emerging patterns/categories/themes, possible networks, related theory, conceptual drawings, future directions, problems, personal/ethical dilemmas.
- Advice: prioritise memoing (write when an idea strikes, no self-censoring), keep memos findable, do not standardise the format; visual memos are possible.

Software for coding (MAXQDA / QDAS)

- QDAS helps to **organise** (many data formats), **analyse** (code, retrieve, visualise, search) and **present** (export, reports, transparency). Myths: the software analyses the data for you (no), you must learn everything in it (no), you must use every feature (no).

- Deductive approach: create codes first, then apply by drag and drop. Inductive approach: create codes directly from the text. **Memos** (any size, attached anywhere) vs. **comments** (only on coded segments, 225-character limit). Retrieval: simple and complex (e.g. nearness of codes). Teamwork: merge projects / export-import.

The Gioia method (Gioia et al., 2013) – a structured inductive approach

Step	Description
I. 1st-order analysis	**Informant-centric**, in-vivo terms (participants' exact words) as codes – similar to open coding
II. 2nd-order analysis	**Researcher-centric** concepts, themes and dimensions – similar to axial and selective coding; think at both levels simultaneously: "What's going on here?" theoretically
III. Data structure	Visual map showing progression from raw data (1st-order concepts → 2nd-order themes → aggregate dimensions); demonstrates rigour (Corley & Gioia, 2004)
IV. Build theory	Specify (temporal) connections between theoretical concepts to build a grounded theory, (process) model or propositions

Rigor, validity and reliability

- Stay close to the data and establish a **chain of evidence** from raw data to abstract concepts and themes; use a **codebook** (write a definition and instructions for when (not) to use each code; consult it when reusing codes); use **memos**; involve **other coders** (multiple people code, compare and resolve differences; done systematically = **intercoder reliability/agreement**); **triangulate** with multiple data sources (Yin, 2009).

Reporting

- **Methods section** should include: justification of the qualitative approach for the RQ; description and justification of the research design (sampling, context); data collection process and sources (preferably a **data table**: type, amount, source, dates, use in analysis; interview topics); data analysis process and methods (preferably with coding scheme and/or data structure; coding steps and sensemaking steps); description of the sensemaking process leading to the main findings.
- **Findings section:** insights from your data; typically **no reference to literature**. Build a logical structure with visuals (e.g. explain a conceptual model/CLD step by step); make clear, unambiguous statements (relationships between concepts); provide ample, **triangulated empirical evidence** (quotes, document passages; tables for additional quotes) so an outside reader can verify. In the text: (1) explain the model/key relationships, (2) explain and illustrate key concepts and causal relationships with quotes, (3) triangulate with several data sources (e.g. interview quotes plus emails and observations).
- **Discussion:** reflect on findings in relation to the RQ and existing literature: how did the findings answer the RQ ("our findings show that/how…")? what is new / what do they contribute to existing work ("while existing work has primarily focused on…, this study demonstrates…"; "our findings extend prior work by…")?
- Practice makes perfect: coding, memoing, sensemaking, modelling/presenting findings.

Lecture 4.1 – Research Project and Writing Up (Quantitative Research)

Prof.dr. Jan de Jonge (Human Performance Management group), 24 September 2026. Bell et al. (2022): Ch. 2 Business research strategies [refresh], Ch. 4 Planning a quantitative research project and research questions, Ch. 7 Writing up quantitative business research.

Qualitative vs. quantitative research

- **Qualitative research** = understand concepts, thoughts or experiences; usually expressed in **words and meanings**; in-depth insights on topics that are not well understood.
- **Quantitative research** = collecting and analysing **numerical data**; used to find patterns, trends, averages, explanations, predictions and **cause-effect relations between variables**; usually based on a fair amount of data. In short: numbers and statistics vs. words and meanings.
- Planning: L4.1 research project and writing up (24 Sept), L5.1 literature review and quantitative research designs (29 Sept), L5.2 nature of quantitative research and questionnaires (1 Oct), L6.1 statistics tutorial (SPSS) (6 Oct), L6.2 secondary analysis and sampling (8 Oct), optional Q&A (13 Oct). **Exam material:** book chapters per lecture (L4.1: Ch. 2, 4, 7; L5.1: Ch. 3, 5; L5.2: Ch. 8, 11, 12; L6.1: Ch. 15, 16; L6.2: Ch. 9, 14) **plus all PowerPoint slides**. How to prepare: read chapters globally before the lecture, in depth after (with the handout), ideally a third time; read checklists/key points and try the review questions.

Chapter 2 – Science and research strategies

- **Science** = structured activities aimed at **observing, understanding, predicting and controlling** certain phenomena; empirical focus on systematic testing using data. In organisations: a scientific approach to systematically solve organisational problems. (Terry Pratchett: science is a method for asking awkward questions and subjecting them to a reality check.)

Basic (theoretical) research	Applied (practical) research
Also pure or fundamental research; problem-solving of a theoretical nature; little impact on action, performance or policy decisions; mainly at universities/institutes (increasingly at companies such as Meta, Google, Microsoft)	Practical problem-solving for a business or management issue; applies theoretical notions (evidence-based solutions); mainly at companies, also universities

- **Research philosophy – central dilemma:** can we know the world objectively, or is knowledge always a subjective representation of reality? (objectivism/positivism vs. constructivism/interpretivism, see L1.1).
- **Two cycles** (see L1.1): problem-solving cycle (key question: how to make it work?) and **empirical cycle** (key question: how to explain it?). **This lecture series focuses on the empirical cycle.**
- **Induction** = a general conclusion formed from a limited set of specific cases (observations); true until now (three white swans → all swans are white). **Deduction** = the conclusion necessarily follows from the premises (swans are white + two swans in the park → these two swans are white).
- **Empirical cycle with swans:** theory "all swans are white" → hypothesis "if we notice a swan, it is white" → testing. Willem de Vlamingh found the black swan in Perth (Dec 1697) and invalidated an accepted "truth" → theory had to be strongly adjusted (falsification).

Theory, model and hypothesis

Theory = "a set of systematically interrelated **concepts, definitions and propositions** that are advanced to **explain or predict phenomena (facts)**." Bell et al. (2022): "a way of explaining observed patterns of associations between phenomena" / "background literature in an area of social enquiry". A theory narrows the range of facts under study, summarises what is known, suggests research approaches, can be empirically tested and can predict new facts.

- **Hierarchy of scientific knowledge:** meta-theory/meta-paradigm (global perspective of how a discipline views reality, e.g. psychological phenomena such as mind and behaviour) → paradigm or **grand theory** (general system of concepts and propositions: biological, cognitive, behavioural, humanistic approach) → **middle-range theories** (Bell; testable, most used in business research).

- **Building blocks of theory:** model, constructs, concepts, variables, hypotheses, definitions, propositions.
- **Model** = schematic representation of reality using analogy; functions: visualisation, simplification, explanation, representation, heuristic. A model representing a theory = **theoretical model**.
- **Variable** = characteristic, trait or attribute that is measured and able to vary; operationalisation of a concept at the empirical level; dichotomous (e.g. gender male = 0 / female = 1; nurse = 0 / physician = 1), continuous (age), or categorical with discrete values (education 1 = primary … 4 = higher; car brand 1 = Volvo … 4 = Toyota; task performance 1-7).

Types of variables

Variable	Meaning
Independent variable (IV)	Cause, stimulus, predictor, input (e.g. workload)
Dependent variable (DV)	Effect, response, criterion, output/outcome (e.g. performance); its value depends on the IV; relation can be negative (−) or positive (+)
Moderating variable	The relation between two variables is **contingent on a third variable** (e.g. relation between salary and satisfaction depends on profession or gender)
Mediating variable	Explains (part of) the relation between IV and DV (e.g. money stress between salary and satisfaction). **Partial mediation:** IV-DV relation still significant; **full mediation:** IV-DV relation no longer significant
Confounding variable	A third variable creating a (spurious) relation (e.g. temperature confounds the positive relation between ice cream sales and child drownings)

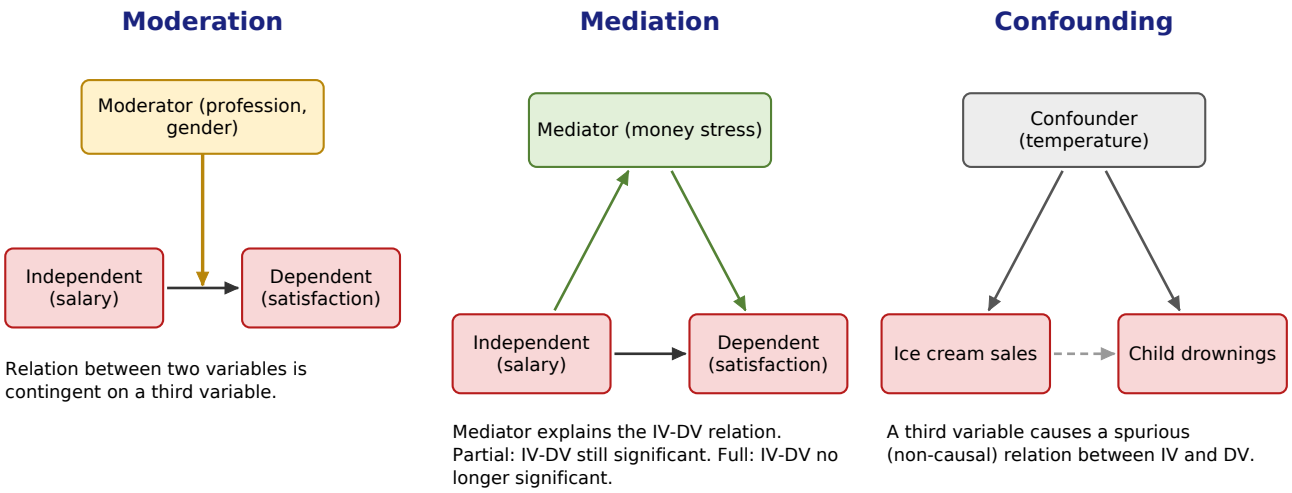


Figure – Moderator (relation depends on a third variable), mediator (explains the relation; partial vs. full mediation) and confounder (spurious relation).

Hypotheses

Hypothesis = a statement formulated for **empirical testing**, usually an expectation following from theory (or previous observation); gives direction to the study (to be proved or disproved).

- **Types: descriptive** (existence, size, form or distribution of a variable), **relational** (relation between two or more variables), **categorical** (differences between groups/classes).
- **A good hypothesis:** adequate for its purpose, uniform and concrete, **testable (!)**, better than its rivals.

Chapter 4 – Planning the project and research questions

- **Advice for a quantitative project:** double check and follow the requirements (e.g. Assignment 2); think about your project; **start early**; be transparent, clear and cooperative to all stakeholders; use resources (supervisor, mentor, fellow students); it is not a bid for a Nobel Prize; ask feedback before, during and after the project.

- **Time management:** realistic timetable, e.g. **Gantt chart**; include data collection time, supervision time, examination time and revision time; plan breaks and days off.
- **Research questions** focus on what you want to know; type depends on the research strategy; can be global in the beginning and more concrete later; should end with a question mark. They guide the literature study, hypotheses, research design, data collection methods, data analysis and write-up.

Criteria for research questions: clarity and non-trivial; researchable; link with theory and research; logically and conceptually linked to each other; neither too broad nor too narrow; contribution to new knowledge; relevant for the field; **KISS** principle (keep it stupidly simple).

Research question hierarchy (train delay example)

Level	Example
1. Management dilemma	Trains are delayed
2. Management question	How can we reduce or eliminate this?
3. Research question	Possible causes: role of passengers, platforms, ICT, staff shortage, locomotives in causing delays? (1-4 are the **independent variables**: passengers' entry behaviour, platform/train entry facilities, malfunctioning of trains/switches/signals/accidents, lack of staff)
4. Investigative questions	Does delay occur due to entry behaviour? Maintenance status of locomotives? Platform layout? ICT problems? Safety issues?
5. Measurement questions	What should be asked or observed to obtain the information?
6. Decision	What is the recommended course of action?

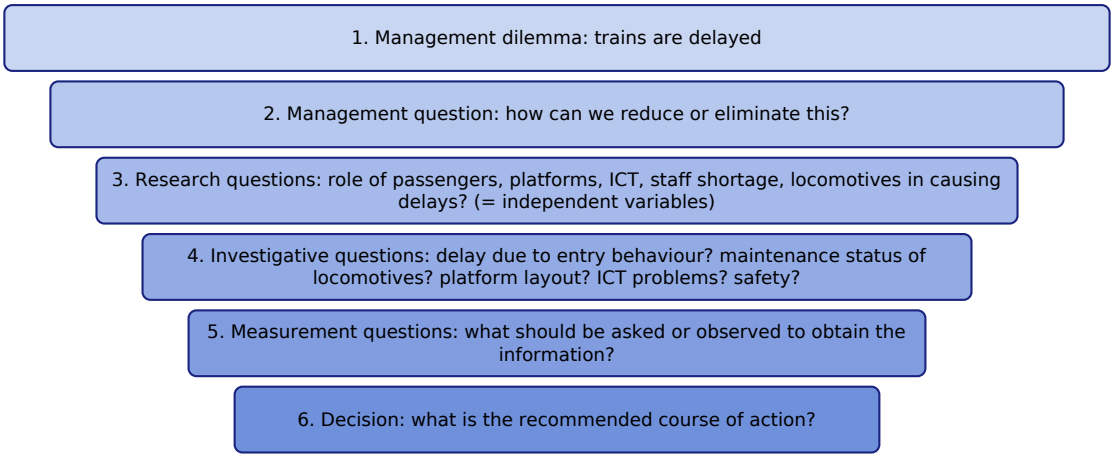


Figure – Research question hierarchy (train-delay example): from management dilemma to decision.

Chapter 7 – Academic writing

- Advice: follow requirements and instructions; start early; use resources and ask for feedback; structure writing and be concise; be persuasive and clear; avoid discriminatory language; **avoid plagiarism and fraudulent behaviour**; also valid for use of generative AI.

Common problems in academic writing (also for Assignment 2): absent macrostructure (organisation and flow); amateur style and tone; no theory or theoretical model; insufficient or lacking definitions; **concepts and operationalisation not in alignment**; insufficient rationale for the research design; inadequate or missing research design; over-engineering of research methods; **no study limitations mentioned**; conclusions not in alignment.

Lecture 5.1 – Assignment 2, Literature Review and Quantitative Research Designs (Part 1)

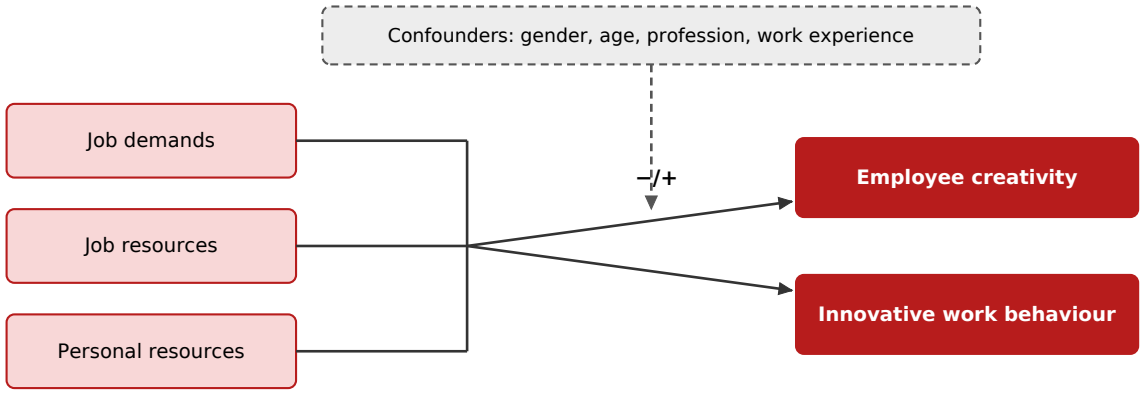
Prof.dr. Jan de Jonge, 29 September 2026. Bell et al. (2022): Ch. 5 Literature review, Ch. 3 Research designs (part 1).

Assignment 2 – "Work redesign" at the Emergency Department

- **Aim:** test which **job and/or personal factors** are associated with healthcare workers' **creativity and innovative work behaviour**. Your role: a team of quantitative researchers (Department of Organizational Behavior) at a Dutch university; a colleague observed that ED staff of a large general hospital were not very creative/innovative in stressful situations; you want to know why.
- **Theoretical frameworks: Job-Job (J-J) fit theory** and **Person-Job (P-J) fit theory**.

Fit = correspondence or match of two constructs. **Supplementary fit:** one characteristic could replace another (identical in function/goal). **Complementary fit:** one characteristic is added to another; includes **Person-Job fit** (compatibility between personal characteristics and job characteristics, e.g. cognitive resilience vs. cognitive job demands) and **Job-Job fit** (compatibility between different aspects of job redesign, e.g. cognitive job demands and cognitive job resources). "Head / heart / hand" (cognitive, emotional, physical) – matches are expected within the same type.

- **Potential questions:** primary question: are the measurement instruments valid and reliable? Secondary: (1) which work and/or personal factors are associated with creativity and/or innovative work behaviour; (2) is there a match; (3) do group differences (profession, gender) exist in creativity/IWB; (4) do group differences exist in the relations of question 1?
- **Research design:** cross-sectional survey with online self-completion questionnaires (**primary data**). **Sample:** 304 of 401 healthcare workers (**response rate 76%**); 143 physicians (47%) and 161 nurses (53%); 155 women (51%) and 149 men (49%); mean age 39.3 (SD 9.4); mean work experience 12.0 years (SD 10.7).
- **Components of the assignment:** (1) determine which factors predict creativity and/or innovative work behaviour by analysing data: **psychometric tests (validity/reliability)**, test associations between variables, test group differences; (2) write a **scientific report**, upload to Canvas by **Sunday 25 October 2026, 23:59** (pdf, one attempt; upload twice: for 1ZV60 and P&PD).
- **Questionnaire:** general questions (profession, age, gender, work experience), **job demands, job resources, personal resources, employee creativity, innovative work behaviour**. Items are in the SPSS database on Canvas with the questionnaire and code book; from the items you create **scales**.
- **Choices:** it is impossible to examine all relations; make well-grounded choices; **KISS** (keep it stupidly simple). General model: job demands, job resources, personal resources (−/+) → creativity, innovative work behaviour, with **confounders** (gender, age, profession, work experience). Subgroup analysis: profession/gender/age groups as moderators of the IV-DV relation; or demographics as predictors (e.g. gender differences in job demands/resources; age-group differences in creativity).



Job-Job fit: demands × resources. Person-Job fit: demands × personal resources (e.g. cognitive resilience).

Figure – General model for Assignment 2: job demands, job resources and personal resources (−/+) predict employee creativity and innovative work behaviour, controlling for confounders.

- **Example hypotheses:** higher cognitive job demands are associated with higher creativity; cognitive job resources are positively related to innovative work behaviour; cognitive resilience (personal resource) is positively related to creativity; a match between demands, resources and outcomes is more likely than non-matches.

Scientific report

Section	Content
Abstract and keywords	Short summary (5 points)
Introduction (20)	Purpose and importance (basic or applied relevance); brief review of prior research and theory; "**funnel reasoning**" ending with specific hypotheses
Methods (20)	Design and procedure, participants, sampling, measures, statistical analysis; reader should be able to **replicate**; mention ethical approval
Results (20)	Findings in chronological order; statistical analyses; tables and figures; **facts rather than interpretations** (SPSS tutorial)
Discussion (20)	Summary of important findings and (non-)confirmation of hypotheses; comparison with previous studies; added value; strengths, limitations, future research, practical implications; answers the research question(s)
References (5), overall flow/effort/writing style (10)	Total 100 points; Title page and appendices (optional)

Chapter 5 – Literature review

- A literature review helps you: know what is already known (not reinventing the wheel); revise and refine RQs/hypotheses; decide how to collect data (design), which measures, how to analyse; increase credibility.
- **Scientific literature** = scholarly, peer-reviewed publications reporting original (empirical) findings and theoretical work. Progress is a continuous accumulation of knowledge ("**cathedral of knowledge**": bricks, walls, foundations of new chapels).
- **Searching:** academic databases (ABI/INFORM, EconLit, EBSCO, SSCI, PsycLIT/PsycARTICLES, Google Scholar, ResearchGate), GenAI, working paper databases, publishers' and journals' websites, authors' websites. **Guidelines:** (1) basic search on topic or author, (2) refine (period, journal quality, article type, language), (3) assess the information, (4) synthesise. Most papers have a **DOI**; TU/e VPN gives access.
- **Relevance check:** read title → abstract → keywords → scan empirical part → read the beginning of the discussion → check the reference list ("cited ref search").
- **Primary sources** = full-text original theoretical or empirical studies; **secondary sources** = compilations (books, systematic reviews).

Journal impact factor (JIF) = average number of citations in a year to articles a journal published in the previous two years. Example: 1000 citations in 2020 to articles published 2018-2019 / 250 articles = **4.00**.

Types of review

Review	Description
Narrative (traditional)	Comprehensive and critical search of current knowledge; identify patterns and trends: "reading and summarising"
Systematic	Replicable, scientific, transparent process minimising bias through (almost) exhaustive searches; often (not always) uses statistics: "reading, analysing, summarising"
Integrative (Torraco, 2016)	Critiques and synthesises representative literature to generate **new frameworks and perspectives**; resolves inconsistencies; more methodological than narrative, less procedural/statistical than systematic: "reading, synthesising, generating"

- **Meta-analysis** = "analysis of analyses": combines results of several related studies to estimate an average or common effect using **effect size statistics**; often part of a systematic review. **Effect size** = strength of relation between two variables on a numeric scale (standardised mean difference, correlation coefficient, odds ratio). Example: nine studies on maternal alcohol consumption in pregnancy and childhood leukaemia: pooled odds ratio 1.56 → +56% chance on average.
- **Critical reviews:** book reviews; **peer review** (anonymous expert peers review a submitted manuscript; editor decides; can take weeks to years).
- **Citing and referencing:** always cite documents you used; **avoid plagiarism** (taking someone's work or ideas as your own; also for GenAI); use quotes with page numbers for literal sentences; citing = in-text, referencing = list at the end (APA preferred; Harvard, Vancouver, Oxford exist; be consistent).

Chapter 3 (part 1) – Research designs

Research design = a framework or blueprint (generalises from sample to population). **Research method** = a technique for collecting data (a specific instrument, e.g. a self-administered questionnaire).

- Example (train delays): RQ "What is the role of passengers in train delays?" → design: cross-sectional survey with fixed questions and answers; sample: passengers at Eindhoven Station in the 3rd/4th week of December 2026; analysis: correlation or regression (testing associations).
- **Key types:** cross-sectional, longitudinal, case study, comparative, experimental (continued in L6.1). **Cross-sectional design** (see L6.1): many cases (> 250), single point in time, ≥ 2 variables, no manipulation, individual differences recorded.
- The **Milgram experiment** (obedience to authority) was shown as an ethical discussion case.

Lecture 6.1 – Research Designs, Questionnaires and Measurement (Part 1)

Prof.dr. Jan de Jonge, 6 October 2026. Bell et al. (2022): Ch. 3 Research designs (continued), Ch. 11 Self-completion questionnaires, Ch. 12 Asking questions.

Chapter 3 – Key types of research design

Key types: cross-sectional (survey), longitudinal (survey), case study, comparative, experimental design.

Cross-sectional design (social survey design)

- Collection of **quantitative data on more than one case** (usually many: > 250 people) at a **single point in time** ("snapshot approach").
- At least two variables (usually many more); tests **relations between variables** (usually via hypotheses); **no manipulation**; individual differences (variations) are recorded (e.g. workload vs. burnout).

Longitudinal design

- Same characteristics as cross-sectional but at **more than one point in time** ("video approach"). Records **intra-individual (within-person)** and **inter-individual (between-person)** differences.
- Tests change in variables (change = Y1 − Y0; Y1 = Y0 + ... + e) and **allows causal inference** (Y1 = Y0 + X0 + ... + e).
- Two key types: **panel study** = repeated measures from the **same sample** at different times; **prospective cohort study** = repeated measures from a cohort (participants sharing a common characteristic such as occupation or demographics) over a long time (often many years); usually different people are included.
- Prospective: exposure → outcome. Retrospective: outcome → exposure.

Longitudinal design vs. causal relations

- **Symmetrical relation:** two variables fluctuate together but a change in one does not cause a change in the other. **Asymmetrical relation:** a change in X is responsible for a change in Y. **Reciprocal relation:** X and Y mutually influence/reinforce each other.
- **Correlation does not imply causation.** Example: effect of daylight (IV) on plant growth (DV): three conditions (365 days daily daylight / every other day / no daylight; same nutrition and water).

Four types of evidence needed for causality: (1) theoretical support, (2) systematic covariation, (3) temporal sequence, (4) non-spurious covariance. If these are not met: no claim of causality.

- **Spurious covariation / confounder:** more firemen ↔ more fires (confounder: cities); more stork nests ↔ more babies (confounder: countryside). Some significant correlations are purely accidental (millions of correlations among thousands of variables; tylervigen.com examples). **We can never prove causation; we can only rule out other causes and explanations.**

Case study design

- **Detailed and intensive analysis of a single case**; deals with complexity and nature of the case; focus on a bounded system with a purpose and functioning parts. Case = a person, an organisation or location, or a single event (e.g. space shuttle disaster).
- **Multiple-case study:** comparison of more cases, focus on individual cases and their unique context. **Longitudinal case study:** follows case(s) over a long period; archival information; retrospective interviewing.

Comparative design

- Using more or less **identical methods on two or more contrasting cases** (also across countries/cultures). Aims: seek explanations for similarities and differences; greater awareness and deeper understanding of social reality in different contexts.

Experimental design – the "gold standard"

Requirements: (1) **random assignment** of people to experimental or control conditions (randomisation); (2) **manipulation of the independent variable**; (3) a **control (comparison) group** (no-treatment condition).

- Other ways to make groups equal: **matching** (e.g. equal distribution of age/gender over groups), **weighting** (re-weight data when one group is under-/overrepresented), **stratification** (first divide into strata, then randomly assign).
- Notation of designs (R = random assignment, O = observation, X = treatment): posttest-only randomised experiment; pretest-posttest; nonequivalent groups (quasi-experiment).

Posttest-only randomised experiment

R	X	O
R		O

Pretest-posttest randomised experiment

R	O	X	O
R	O		O

Nonequivalent groups (quasi-experiment)

N	O	X	O
N	O		O

Solomon four-group design (true experiment)

R	O	X	O
R	O		O
R		X	O
R			O

R = random assignment N = non-random assignment O = observation/measurement X = treatment/manipulation

Each row is one group: experimental groups receive X; control groups do not. Columns are time points (left to right).

Figure – Notation of experimental designs (R = random assignment, N = non-random, O = observation, X = treatment) including the Solomon four-group design.

- **Field experiment:** natural, real-life setting; participants usually do not know they are studied. **Laboratory experiment:** in a lab or contrived setting (e.g. office simulation); participants usually know they are studied.

Validity in experimentation

- **Validity** = degree to which a study measures what it is supposed to measure / can scientifically answer its question. **Reliability** = extent to which a measurement gives consistent results. Validity ≠ reliability.

Type	Question
Internal validity	Do the conclusions about a demonstrated relation truly imply "cause"? Are results attributable to the independent variable and not to a rival explanation?
External validity	Does an observed causal relation generalise across people, settings, environments and/or times? (generally about population)
Measurement validity	Does a measure capture the phenomenon it is intended to measure?
Ecological validity	Do findings/instruments capture daily-life conditions, opinions, values and attitudes in the natural habitat? Subtype of external validity: generalisation to real-life settings.

- Internal and external validity are **reciprocally related**: controlling internal validity (highly controlled environment) decreases external validity; controlling external validity (more realistic setting) decreases internal validity.

Threats to internal validity

Threat	Meaning
History	Events during the experiment confuse the relation (e.g. a merger)
Maturation	Subjects change over time (know more, become stronger)
Testing	Taking a test affects scores on a second test
Instrumentation	Changes in the measuring instrument or the observer
Selection	Differential selection of subjects for experimental and control group (e.g. field experiments)
Statistical regression	Regression to the mean: high scores tend to decrease, low scores to increase
Experiment mortality	Group composition changes during the test (loss of subjects, attrition)

Quasi-experiments and other experimental designs

- **Quasi-experiment:** used when requirements of a true experiment cannot be met (no randomisation, no controlled manipulation e.g. natural event like a new way of working, no real control group). Prominent in evaluation and intervention research (example: Niks et al., 2018).
- **Randomised block design:** reduce experimental error by grouping units into blocks (variability within blocks minimised, between blocks maximised) and randomly assigning treatments within each block. Blocking factors: weight, income, SES, occupational level.
- **Latin square design:** two major extraneous factors (type of treatments and their order); same number of columns, rows and treatments; each treatment appears once in each row and column.
- **Ex-post facto design:** "after the fact"; pre-existing qualities of groups are compared on a dependent variable without random assignment (quasi-experimental). Example: weight (under/normal/overweight) and self-esteem.
- **Factorial design:** more than one independent variable (e.g. 2×3 factorial design).
- **Solomon four-group design ("platinum design"):** true experimental design with random assignment combining pretest-posttest-with-control-group and posttest-only-with-control-group; controls for all kinds of extraneous variation. The Milgram experiment was also discussed (ethics).

Chapter 11 – Self-completion questionnaires

- **Self-administered questionnaire** = data-collection approach using **written communication** (text, not observation); also called self-completion questionnaire or "survey"; involves a sample of people; respondents answer written questions that are recorded for analysis.
- Aim: collect, analyse and generalise sample results to a larger population **quickly and relatively cheaply**. Standard tool since the 1930s; electronic surveys increased popularity.
- Method choice depends on: costs and effort, research topic, response time, accessibility and spread of population, **anonymity or confidentiality** (anonymity = nobody, not even the researcher, can identify the respondent; confidentiality = researcher knows identity but does not disclose it).

Pros	Cons
Relatively cheap; quick to administer; no interviewer effects; standardised (no interviewer variability); convenient for respondents	No assistance or elaboration possible; cannot ask too many/too difficult questions; not knowing who exactly responded; missing data and low response rates

Questionnaire bias

Questionnaire bias = the tendency of a sample statistic to **systematically over- or under-estimate a population parameter**. Problem: you do not get accurate data that reflect the population.

Bias	Description / remedy
Non-response bias	Answers of participants differ systematically from non-participants (e.g. highly loaded teachers do not participate in a workload study). Remedies: reminders/call-backs, substituting another person, incentives, creating a non-response sample and weighting afterwards, Missing Value Analysis (MVA).
Voluntary response bias	People choose themselves to participate (often on controversial topics); over-represents strong opinions; samples are invalid and cannot be recovered.
Social desirability bias	Answering in a way viewed favourably by others: over-reporting desirable, under-reporting undesirable behaviour.
Extreme-scoring tendency	Always scoring extremes (e.g. 1 or 5). Avoid by using both negatively and positively worded items, not-too-extreme statements, post-hoc extreme-scoring analysis.
Positive and negative affect	Respondent's mood affects evaluations (positive: cheerful, proud, enthusiastic; negative: sad, fearful, distressed). Remedy: add mood scales and **control for them statistically** (partialling out).
Sampling bias	Due to sampling procedures (see Lecture 6.2).

- Combined example: TU/e survey (Sept 2021) reported > 90% of students fully vaccinated, but response rate was only 41% – non-vaccinated students may not have participated (non-response bias) and the survey over-represents vaccinated (voluntary response bias).

Web-based surveys, diaries, ESM

- **Web survey types:** target web surveys (researcher controls who is asked), self-selected surveys (pop-up, little researcher control), social media-based surveys. Pros: short turnaround, visual stimuli, many surveys over time, many respondents, often anonymous, flexible analysis. Cons: recruiting the right sample is costly and time-consuming, browser problems, web-conversion expensive, IP-address checks privacy-sensitive.
- **Diaries (researcher-driven):** participants record their own behaviour during a specific period (week, 10 days, month), usually a brief structured questionnaire, filled out several times a day. Pros: more precise estimates, information on sequencing (longitudinal). Cons: more expensive and time-consuming, higher attrition, respondents become less diligent.
- **Experience / event sampling method (ESM):** variant of diary studies capturing transient affective states; participants are prompted (usually smartphone) at specific times or after specific events; immediate data collection, less memory distortion.

Chapter 12 – Asking questions

- Question formulation determines quality/usefulness of data. First consideration: **open-ended versus closed questions**. Question types: personal (gender, age, education), about others, informant questions, work-related (workload, leadership), attitudes, beliefs, norms and values, knowledge and skills.
- **Scaling** = assigning numbers (or symbols) to a property of objects so that properties of numbers apply. Depends on study objective, response form, degree of preference, data properties, number of dimensions, scale construction.

Hierarchy of scales

Scale	Characteristics	Statistics	Example
Nominal	Classification only (mutually exclusive, collectively exhaustive); no order	Mode	Gender
Ordinal	Classification + order (not how much greater/less)	Mode, median	Education (primary, secondary, higher)
Interval	Classification + order + equal distance; arbitrary zero	Mode, median, mean	Temperature (Celsius)
Ratio	Classification + order + distance + absolute zero	Mode, median, mean	Age in years, weight

- Age in years = **ratio**; education as primary/secondary/higher = **ordinal**. A ratio scale can be transformed downward: age → ordinal (0-25, 26-50, 51-75, 76+) → nominal (non-adult 0-17, adult 18+). **Aim for the highest level of the scale hierarchy possible.**

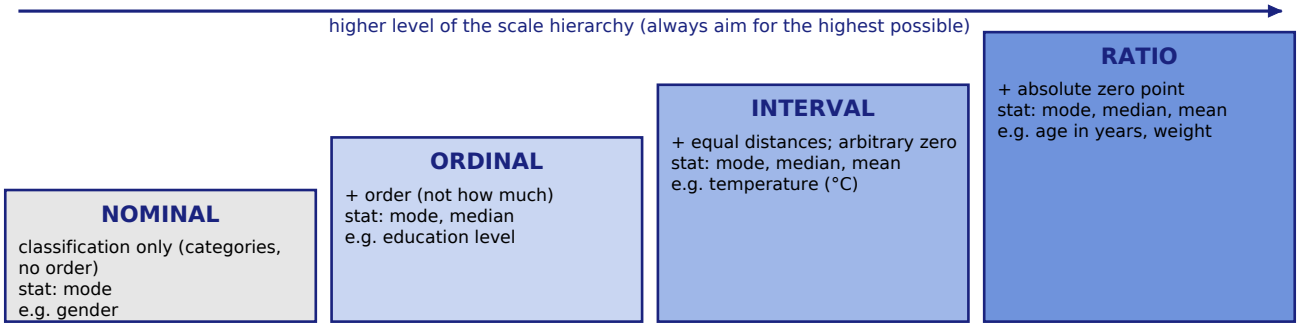


Figure – Hierarchy of measurement scales: nominal → ordinal → interval → ratio (each level adds properties).

- Parametric tests on ordinal measures are strictly speaking incorrect but usually acceptable (small risks).
- **Response methods:** simple category scale, multiple-choice scale, multiple-rating scale, **Guttman scale** (items form a continuum: whoever answers "no" to one item also answers "no" to the following items), **Likert scale**, **semantic differential scale** (bipolar adjectives on 1-5, e.g. angry–calm; emotional stability in Assignment 2), graphic rating scale (e.g. pain scale).
- **General rules for question design:** ask what you want to know and imagine how you would answer yourself; avoid ambiguous, too long, **double-barrelled**, too general, too extreme or leading questions; avoid too many negatives and technical or obscure wording; keep answer categories simple.

Lecture 6.2 – Measurement, Sampling and Secondary Analysis

Prof.dr. Jan de Jonge, 8 October 2026. Bell et al. (2022): Ch. 8 Sound measurement, Ch. 9 Sampling in quantitative research, Ch. 14 Secondary analysis and official statistics.

Chapter 8 – Sound measurement

- Quantitative research attempts to **measure psychosocial phenomena and their associations**.

Concept, construct, variable

Concept (Bell et al., most important): a building block of a theory with a generally accepted collection of meanings or characteristics; the points around which business research is conducted; a label for elements of the social world that share common features; a category for organising ideas and observations.

Construct: a concept that has been **deliberately invented for a special scientific purpose**; used (1) in theoretical schemes and (2) defined so that it can be operationalised and measured.

Concept	Construct
General idea/understanding of a phenomenon	Specific operationalisation of a concept
Broader in scope	More specific in scope
Applied to actual and possible cases	Applied to actual cases only
Makes theoretical claims	Used to measure concepts

- **Variable** = characteristic, trait or attribute that is measured (a numeral can be assigned; it can vary): the operationalisation of a construct at the empirical level; dichotomous (0,1), continuous, or categorical with discrete values.

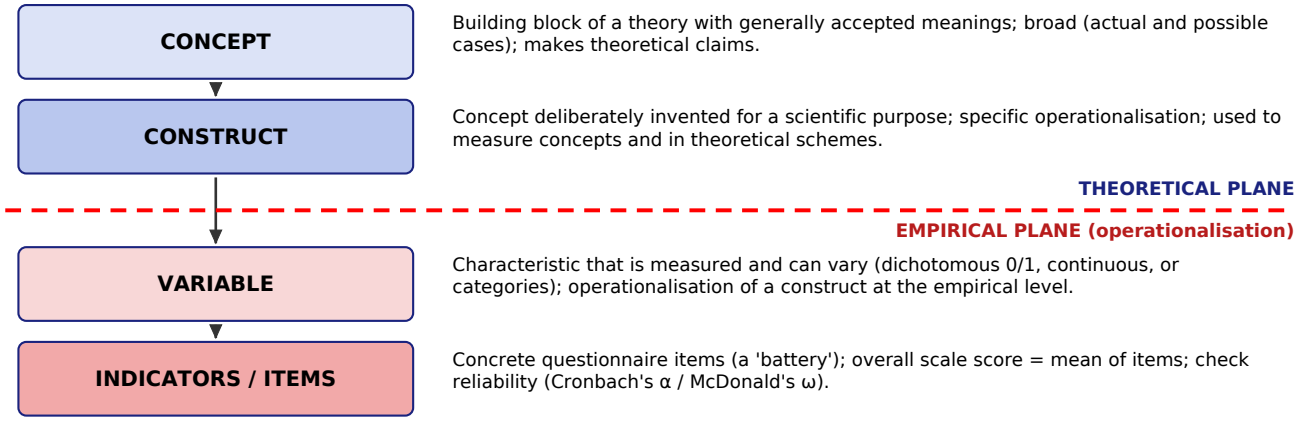


Figure – From concept to construct to variable to indicators/items: theoretical plane vs. empirical plane.

Measurement

- **Measurement** = assignment of numbers to events according to a **mapping rule**; to discover extent, dimensions, quantity or capacity by comparison with a standard (benchmarking). Aim: highest quality, lowest-error data for testing hypotheses.
- Steps: (1) select the empirical event, (2) develop the mapping rule(s), (3) apply the mapping rule(s).
- Researchers measure several **indicators**, operationalised by several concrete **items** (a "battery" of items, e.g. six burnout items; overall score = mean of items, e.g. M = 5.17, SD = 0.37).

Measurement error

- **Systematic error:** caused by a mis-calibrated instrument affecting all measurements. **Random error:** naturally occurring error expected with any measure.
- **Sources:** participant (status, social class, ethnic group), situational factors (presence of others, anonymity violation, location), measurer/interviewer (rewording, reordering), data-collection instrument (complex words, mechanical defects, ambiguous questions).

- **Rating scale errors: leniency / non-leniency** (consistently easy or hard rater = extreme scoring), **central tendency** (reluctance to give extremes), **halo effect** (generalised impression carried over from one rating to others).

Three criteria for sound measurement

- **Validity** (measures what it intends to measure), **reliability** (consistency and stability), **practicality** (economy, convenience, interpretability; Ch. 2 Bell). A measure can be reliable but invalid; it cannot be valid if unreliable.

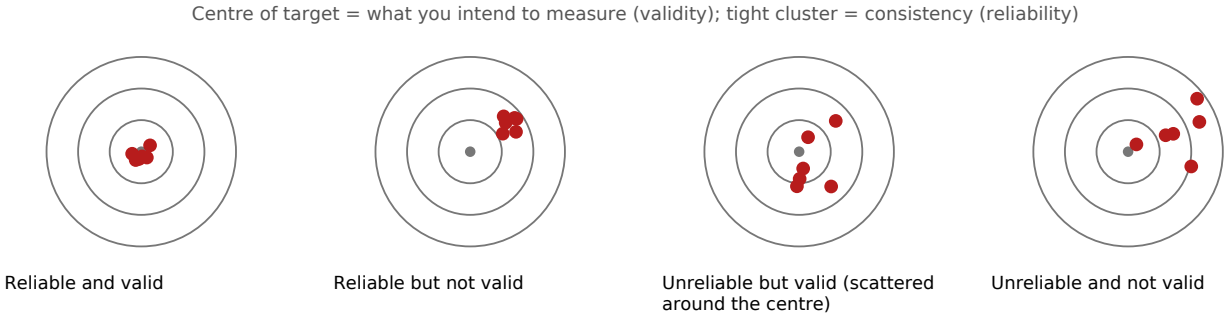


Figure – Validity vs. reliability: reliable = consistent (tight cluster); valid = on target. A measure can be reliable but invalid; it cannot be valid if unreliable.

Reliability

- Types: internal consistency, test-retest, etc. **Test-retest reliability:** same measure at time 1 and time 2; high correlation = high reliability.
- **Cronbach's alpha (α)** (Cronbach, 1951): measures **internal consistency** (a form of reliability) of a multi-item scale; range 0.0-1.0; **alpha is highly sensitive to the number of items**; formula uses n (number of items), sum of individual item variances and total variance of the overall score. Rule: **scales with α ≥ .60 can be used for further analysis**. Only if items are internally consistent may you compute a new latent variable (overall mean score; see Assignment 2).
- **McDonald's (1999) omega (ω):** estimates reliability from the **factor model** (share of total-score variance due to a common factor); uses factor loadings and unique/error variances; rule ω ≥ .60.

	Cronbach's alpha	McDonald's omega
Measurement model	Assumes equal item loadings (tau-equivalent)	Item loadings may differ (congeneric)
Requires	Item covariances	A fitted factor model
If loadings are unequal	Underestimates reliability	Stays accurate
Evaluation	Most commonly reported	More accurate estimate

Validity and factor analysis

- **Factor analysis** reduces many items to fewer factors (latent constructs) by extracting maximum common variance; the factor score can be used instead of the items. Used as evidence for **construct validity**.

Method	Aim	Character
Principal component analysis (PCA)	**Item reduction**	Explorative; uses all variance; successive components explain the largest remaining variation
Principal axis factoring (PAF)	**Factor reproduction**	Confirmative; only shared (common) variance, excludes unique variance

- Reduce items of a newly developed scale → PCA. Replicate the factor structure of an existing scale → PAF. Example: "overall" social support with supervisor, co-worker and family support factors (4 items each).
- **Practicality:** economy (instrument length vs. interview time and costs), convenience (easy to administer, clear instructions, good layout), interpretability (others must be able to interpret results).

Chapter 9 – Sampling

Sampling = the selection process of a **subset of individuals from within a population** selected for investigation, to estimate characteristics of the whole population and draw conclusions for that population.

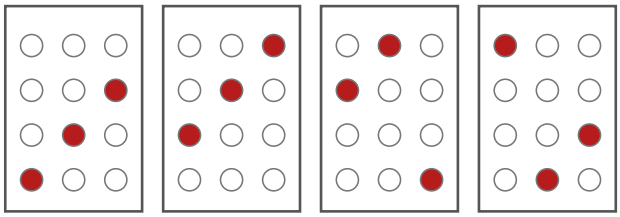
- Reasons: lower costs, lower effort, greater accuracy, greater speed, availability of most population elements.
- **Unit of analysis:** persons, dyads, triads, groups, teams, departments, organisations, institutions, countries. Choice depends on research problem (level of answers), method (level of measurement) and strategy (level of intervention).
- A good sample represents the population. Representativeness = **accuracy** (unbiased; no systematic variance) and **precision** (no random fluctuation/sampling error; measured by the standard error – smaller = more precise).
- **Required sample size** depends on: population size, interest in subgroups, variance in the population, degree of reliability, desired precision, power (strength to detect true effects).

Probability sampling (random selection; equal chance)

Type	Description
Simple random	Known and equal chance of selection
Systematic	Every k-th element
Stratified	Include elements from each segment (gender, class, occupation); few sub-groups, random elements within each; homogeneity within, heterogeneity between sub-groups
Cluster	Population divided in groups; some groups randomly selected (area/street sampling); many sub-groups; heterogeneity within, homogeneity between sub-groups

- **Standard error is smaller with stratified sampling** (variation between strata eliminated) and **larger with cluster sampling**.

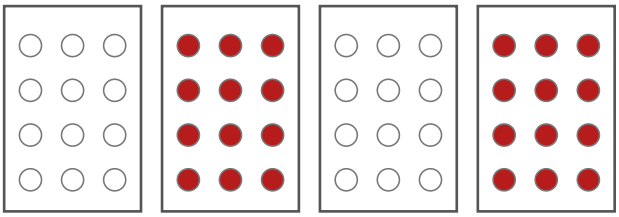
Stratified sampling



Few sub-groups; random elements from EVERY sub-group.

Homogeneous within, heterogeneous between sub-groups → smaller standard error.

Cluster sampling



Many sub-groups; random sub-groups chosen (all/some elements).

Heterogeneous within, homogeneous between sub-groups → larger standard error.

Filled dot = element selected into the sample.

Figure – Stratified sampling (random elements from every sub-group) vs. cluster sampling (random sub-groups).

Non-probability sampling (arbitrary; not equal chance)

- **Convenience** (availability), **snowball** (new participants via previous ones), **quota** (people who fit categories).

Sampling theory

- About 95% of sample means lie within ±1.96 SD (≈ 2) of the population mean. For a probability sample, we are 95% certain the population mean lies in sample mean ± 1.96 × standard error = **95% confidence interval**.
- Example: 495 marathon runners, mean injuries 11.3, SE 3.1 → CI = 11.3 ± (1.96 × 3.1) = 11.3 ± 6.1 → **5.2 to 17.4**. **Smaller SE → narrower CI → more precise estimate**; larger SE → wider CI.

95% confidence interval for the population mean = sample mean ± 1.96 × SE = 11.3 ± 1.96 × 3.1

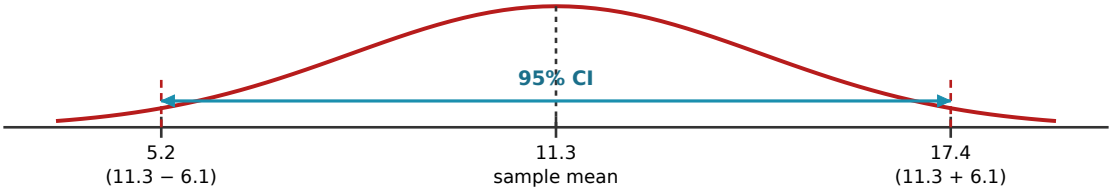


Figure – 95% confidence interval: sample mean 11.3, SE 3.1 → 5.2 to 17.4. A smaller standard error gives a narrower, more precise interval.

Sampling error vs. sampling bias

- **Sampling error:** difference between sample and population even with probability sampling (population specification error or selection error); **decreases as sample size increases**.
- **Sampling bias:** distortion of representativeness: some groups over-represented, others under-represented. Sources: non-probability sampling, inadequate sampling frame, non-response. Larger samples do not fix bias.

Chapter 14 – Secondary analysis and official statistics

Secondary data = information that has **already been collected and recorded by someone else, usually for other purposes**. **Secondary analysis** = (re)analysis by researchers who were probably not involved in the data collection, for purposes probably not foreseen by the original collectors. Examples: European working conditions survey (300+ variables, N ≈ 45,000), tweets, CBS Statline, organisational log files.

Advantages	Disadvantages
Saves time and money; usually high quality; supplements primary data; exploring new areas; subgroup analysis; longitudinal and cross-cultural research; efficient use of data; no ethical clearance necessary	Difficult to find; not all relevant variables/information present; lack of familiarity; not detailed/accurate enough; not specific to needs; can be too complex; no control over data quality

- **Official statistics** (state agencies): data already collected, no respondent reactivity, cross-sectional and longitudinal and cross-cultural analysis possible. Reliability/validity at risk due to time and cultural differences; **ecological fallacy** (inferences about individuals or organisations cannot be drawn from aggregated data). They are **unobtrusive measures** = data collection and analysis do not impinge on the subjects of the study.
- **Data mining** = uncovering knowledge/patterns from databases in data warehouses (valid, novel, useful, understandable patterns); used to explore, discover, predict and test hypotheses. Applications: historical/unidimensional information, trends and patterns (products often bought together). Tools: pattern recognition (stolen credit cards from 12 million daily transactions), prediction tools, risk models (e.g. UK car thefts 2002-2019 and police funding cuts).
- **Big data** = information assets with such high **volume, velocity and variety** that they require specific technology and analytical methods to turn into value; data sets beyond the ability of commonly used software to capture, curate, manage and process within a tolerable time.

Lecture 7.3/7.4 – Exploring Cause-Effect Diagrams

Dr. J.P.M. Wouters (slides Oct 2020), Department of Industrial Engineering & Innovation Sciences, ITEM group. Agenda: recap L7.1, components of a cause-effect diagram, demarcating the problem, research objectives and research questions.

Recap L7.1 – Step 1 of the problem-solving cycle: problem definition

- Problem definition = **search, choose, demarcate and plan**: collect insights internally and externally; synthesise all information; select and demarcate the business problem; decide on research framework (theory and deliverables); plan (when to do what and why, using which resources); write the project proposal.
- **Steps to define a business problem:** intake → external orientation → internal orientation → synthesise information. **These steps are iterative.**

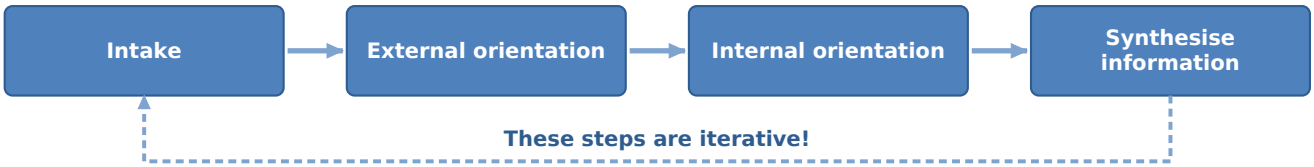


Figure – Steps to define a business problem (intake, external orientation, internal orientation, synthesise information); the steps are iterative.

- **Building blocks scheme (accountability):** table with source (interviews with stakeholders, interviews with experts, management minutes/reports, theory, annual reports, internal data) – findings – concept in the cause-effect diagram. It makes the diagram traceable.

Components of a cause-effect diagram

1. **External triggers** (green) – analysed with **PESTEL** (Political, Economic, Social, Technological, Environmental, Legal) and **Porter**'s micro-environmental forces (customers, distributors, suppliers, competitors). 2. **Root causes** (blue) – internal: organisation structure, culture, strategy, procedures/systems, staff (competencies/motivation), leadership (style), innovation/production/networking capabilities (7S model). 3. **Intermediate (mediating) causes** (orange). 4. **Symptom (= the pain / problem)** (dark red) – the one dependent variable. 5. **System boundary** (red dashed line) – separates external triggers from the organisation's internal causes.

- **Ahrend example:** external triggers (economic crisis, customer trends towards service, new technologies/demand for new products, increased competition) and root causes (outdated systems, inadequate leadership – too product oriented, too little innovative power) lead via mediating causes (less demand, increased workload, decreasing margins, unmotivated staff, customer dissatisfaction, current products less attractive) to the symptom **continuity in danger**.

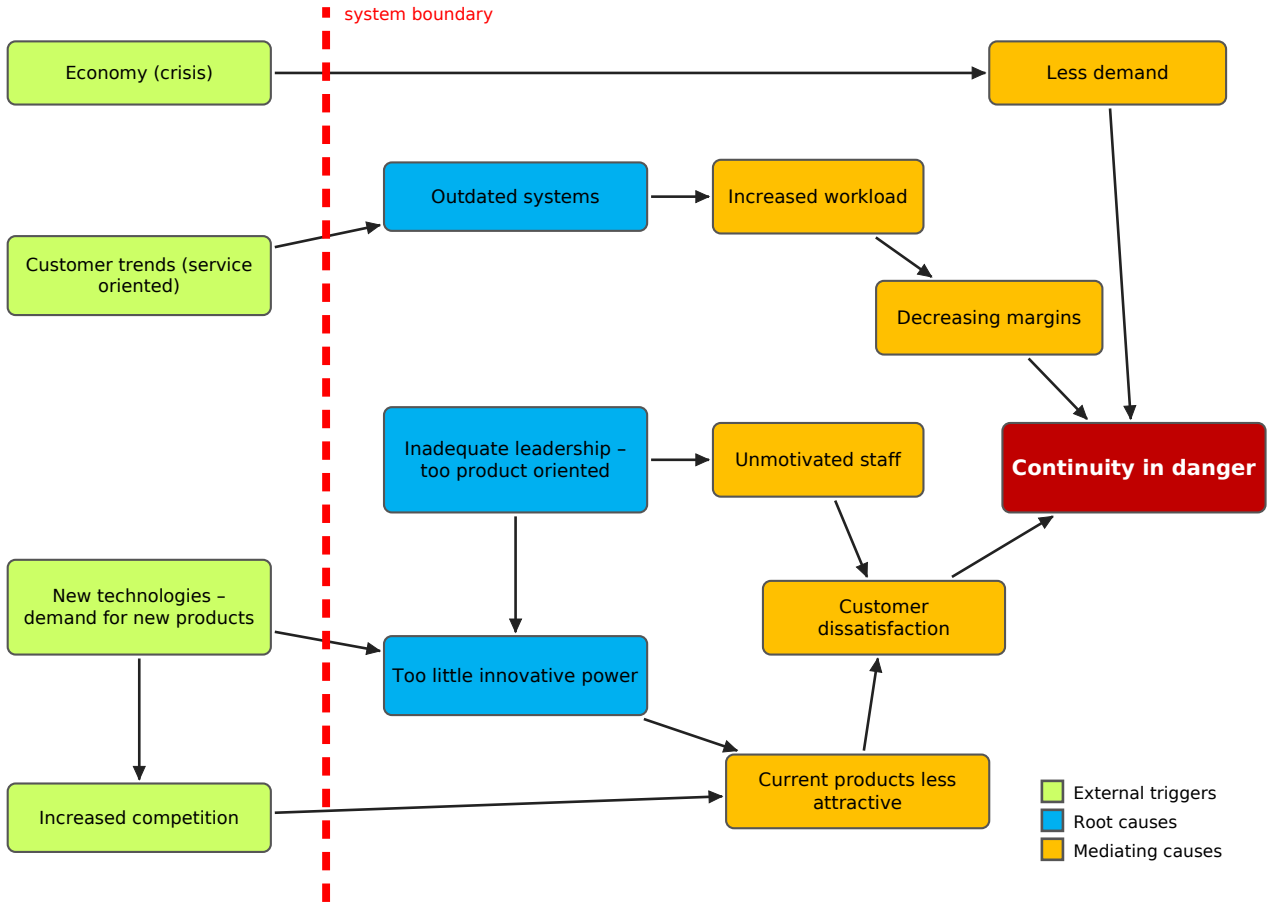


Figure – Cause-effect diagram of the Ahrend case: external triggers (green) and root causes (blue) lead through mediating causes (orange) to the symptom 'continuity in danger' (red); the dashed red line is the system boundary.

Do's for a cause-effect diagram

- **Keep it simple (max. 12/14 concepts)**; go from complex to essential; use it as a **communication tool**.
- **Converge to one dependent variable.**
- Describe in **chains of cause-effect reasoning** (too often it is a process scheme or a historical account – wrong).
- Build and account for it on the basis of the **building blocks scheme**.
- The initial (set-up) diagram is "as good as possible but not yet reliable and complete".

Demarcating the problem

- **Characteristics of business problems:** a business problem is **making a choice** in the cause-effect diagram; **open-ended** (multiple solutions possible); **embedded in a social system** (several stakeholders with different, potentially contradictory perspectives and interests); often bound to limited time and resources.
- Focusing the Ahrend diagram: choose a part of the diagram, each linked to a theory area – leadership and motivational theories (inadequate leadership → unmotivated staff → customer dissatisfaction), operations management theories (outdated systems → increased workload → decreasing margins), innovation management theories (leadership/innovative power → current products less attractive).
- **Criteria to choose the business problem:** relevance (performance relationship), feasibility, corporate supervisor's responsibility area, researchability (availability of data), scientific news value. Then demarcate the system to be investigated.
- **Demarcating problem statements:** be as specific as possible in terms of structure, content, time, place. Vague: "Workload (performance indicator) in production (business process) is too high (unwanted outcome)." Specific: "Workload in production is too high in the Eindhoven factory (place) during the evening shift (time) for older employees (content) working in teams of less than 4 people (structure)."

Research objectives and research questions

- **Assignment wording:** Bachelor thesis = investigate factual causes and consequences of the faulty process (problem) and investigate possible solutions. Master thesis = (re)design the faulty process.
- **Research questions** direct the search for information: what is the relevant theoretical framework; what are the bottlenecks in the empirical context (IST compared to theory); what are possible solutions. RQs often form the structure/titles of the thesis chapters; **methodology must be discussed per RQ**.

Conclusions

(1) Role of theory is still limited in the problem identification phase; theory can be used somewhat when naming and combining phenomena; for demarcation it helps to know which literature areas play a part. (2) **Problem identification is the crucial phase.** (3) It is both an analytical and a socio-political challenge. (4) Determining the business problem is a **joint choice**.